

Python

```
def sum(numbers):
    if len(numbers) == 0: # BASE CASE
        ❶ return 0
    else: # RECURSIVE CASE
        ❷ head = numbers[0]
        ❸ tail = numbers[1:]
        ❹ return head + sum(tail)

nums = [1, 2, 3, 4, 5]
print('The sum of', nums, 'is', sum(nums))
nums = [5, 2, 4, 8]
print('The sum of', nums, 'is', sum(nums))
nums = [1, 10, 100, 1000]
print('The sum of', nums, 'is', sum(nums))
```

And here is the equivalent JavaScript program, *sumHeadTail.html*:

JavaScript

```
<script type="text/javascript">
function sum(numbers) {
    if (numbers.length === 0) { // BASE CASE
        ❶ return 0;
    } else { // RECURSIVE CASE
        ❷ let head = numbers[0];
        ❸ let tail = numbers.slice(1, numbers.length);
        ❹ return head + sum(tail);
    }
}

let nums = [1, 2, 3, 4, 5];
document.write('The sum of ' + nums + ' is ' + sum(nums) + "<br />");
nums = [5, 2, 4, 8];
document.write('The sum of ' + nums + ' is ' + sum(nums) + "<br />");
nums = [1, 10, 100, 1000];
document.write('The sum of ' + nums + ' is ' + sum(nums) + "<br />");
</script>
```

The output of these programs is shown here: